

Knowledge Distillation via Temporal Frequency for Generative Semantic Image Segmentation

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Abstract—Skin lesion segmentation is critical for early melanoma detection, yet the high computational overhead of generative diffusion-based models hinders practical clinical deployment. While knowledge distillation (KD) offers a promising approach toward model compression, existing spatial-domain methods often fail to capture the unique coarse-to-fine denoising dynamics inherent in diffusion processes. To address this limitation, we propose Temporal Frequency Knowledge Distillation (TFKD), a novel timestep-aware framework that operates in the Fourier domain to dynamically align the spectral characteristics of the teacher and student models across the denoising trajectory. By explicitly transferring structural and boundary knowledge according to the spectral progression of different denoising stages, TFKD ensures a lightweight student model robustly retains the generative prior of its larger counterpart. Extensive evaluations on the ISIC2018 and HAM10000 benchmarks demonstrate that TFKD consistently outperforms existing distillation techniques, achieving a superior balance between segmentation accuracy and inference efficiency. Furthermore, generalization experiments on the DUTS dataset reveal that the distilled model remains competitive beyond dermoscopic imaging. These results establish TFKD as a simple, highly effective, and plug-and-play strategy for deploying compact diffusion-based segmentation models in both medical and general vision tasks. Code is available at <https://github.com/hoangdv05/TFKD>.

Index Terms—Medical image segmentation, Skin lesion segmentation, Diffusion-based model, Knowledge distillation

I. INTRODUCTION

Skin lesion segmentation is a key task in medical image analysis, aiming to identify lesion regions from dermoscopic images. Reliable segmentation is clinically important because lesion shape, size, and boundary appearance are closely related to diagnostic cues such as asymmetry and border irregularity [1]. Precise delineation is also useful for downstream analysis, including lesion measurement, disease monitoring, and computer-aided diagnosis. In practice, however, lesion delineation is often performed manually by clinicians or experts, making it time-consuming, labor-intensive, and difficult to scale. This creates a strong need for automated methods that can support fast and reliable lesion analysis.

To address this need, many deep learning methods have been developed for automatic lesion segmentation. CNNs have achieved strong results by learning local patterns and hierarchical visual features [2], [3], while ViT-based models improve long-range dependency modeling and global context

understanding [4], [5]. Despite these advances, dermoscopic lesion segmentation remains difficult because lesion boundaries are often blurry, low-contrast, and visually similar to surrounding healthy skin. Lesions also vary greatly in shape, size, color, and texture, which makes stable generalization across cases challenging. In many difficult samples, the model must distinguish subtle lesion cues from normal skin appearance, and prediction quality can degrade noticeably around the boundary region, which is often the most clinically relevant part.

More recently, DDPMs [6] have emerged as a promising direction for medical image segmentation. Instead of predicting a segmentation mask in a single step, DDPM-based methods formulate segmentation as a progressive denoising process, which is well-suited to recovering complex structures and refining uncertain boundaries. This generative formulation allows the model to improve the prediction gradually rather than making a direct one-step decision, which is particularly helpful when lesion contours are ambiguous or weakly defined. As a result, diffusion-based frameworks such as MedSegDiff [7], DermoSegDiff [8] have shown strong performance in boundary-sensitive segmentation tasks. However, these gains come at a substantial computational cost. Their large model size and iterative multi-step inference make diffusion models difficult to deploy in practical clinical settings where efficiency, memory usage, and inference speed are also important [9], [10].

Knowledge Distillation (KD) [11] offers a practical way to reduce this cost by transferring knowledge from a large teacher model to a compact student model. Existing KD methods usually align teacher and student representations through feature maps, attention maps, or final outputs [12]–[14]. These strategies have been widely used to compress segmentation networks and improve lightweight models. However, most of them operate mainly in the spatial domain and focus on overall representation matching. As a result, they may not fully capture the difference between coarse lesion layout and fine boundary details, although both are important in dermoscopic segmentation. This limitation becomes more apparent in diffusion-based segmentation, where the prediction is formed progressively over multiple timesteps rather than being produced all at once. A student model may therefore

benefit not only from what knowledge is transferred, but also from when and in what form that knowledge is transferred during denoising.

From this perspective, the frequency domain provides a useful alternative for distillation. In intermediate feature representations, low-frequency components generally correspond to more slowly varying coarse-scale patterns, whereas high-frequency components correspond to more rapidly varying fine-scale patterns. Importantly, these feature-space frequencies should not be interpreted as preserving the exact frequency characteristics of the original input image. This view is especially relevant to diffusion models. Recent analysis from a Fourier-space perspective shows that diffusion models tend to recover low-frequency content earlier, while high-frequency details are refined later in the reverse process [15]. This spectral progression suggests that distillation should adapt to the denoising stage rather than remain fixed across timesteps.

Motivated by this observation, we propose a timestep-aware spectral distillation framework that matches the coarse-to-fine denoising dynamics of diffusion models. Instead of applying uniform supervision across all timesteps, our method adapts knowledge transfer to the spectral characteristics of each stage, enabling the student to learn both coarse lesion structure and fine boundary details more effectively.

Our main contributions are summarized as follows:

- Firstly, we introduce **Temporal Frequency Knowledge Distillation**, dubbed TFKD, a timestep-aware frequency-based distillation method for diffusion-based image segmentation.
- Secondly, we demonstrate that the proposed TFKD consistently improves LiteDermoSegDiff [16] on the ISIC2018 [17] and HAM10000 [18] benchmarks.
- Thirdly, we validate TFKD on DUTS [19] in a general setting and show that the proposed strategy remains effective beyond skin lesion segmentation.

The rest of the paper is organized as follows. Section II reviews related work. Section III presents the proposed method. Section IV describes the experimental setup and reports the main results. Section V provides ablation studies, and Section VI finally concludes the paper.

II. RELATED WORK

A. Medical Image Segmentation

Medical image segmentation has been shaped primarily by encoder-decoder convolutional architectures. U-Net [2] established the basic design, while later variants such as UNet++ [20] and UNet 3+ [21] improved multi-scale fusion and skip connectivity for better spatial detail preservation. To address the limited global receptive field of CNNs, transformer-based models were introduced into medical segmentation, with representative examples including TransUNet [4] and Swin-Unet [5]. In skin lesion segmentation, hybrid and boundary-aware models such as FAT-Net [22], Boundary-Aware Transformers [23], and more recent boundary-focused transformer variants such as BDFormer [24] have further emphasized contour

recovery. Despite these advances, skin lesion segmentation remains difficult because dermoscopic images often exhibit low contrast, ambiguous borders, inter-observer variation, and artifacts such as hair and specular reflections.

Diffusion-based segmentation has recently emerged as an alternative to one-shot discriminative prediction. MedSegDiff [7] was among the first general medical segmentation frameworks to model segmentation as a denoising process, introducing dynamic conditional encoding and the Feature Frequency Parser for multi-scale refinement. MedSegDiff-V2 [25] further strengthened this line with a transformer-based backbone. In dermatology, DermoSegDiff [8] showed that diffusion-based lesion segmentation can better handle uncertain and boundary-sensitive cases through iterative refinement. However, these gains typically come with large models and repeated denoising steps, which increase both memory usage and inference time.

B. Knowledge Distillation

Knowledge Distillation has become a standard strategy for compressing large teacher models into lightweight student networks. Classical distillation methods mainly rely on softened output matching [11], intermediate feature imitation [13], and attention transfer [14]. These ideas were later extended to semantic segmentation, where both channel-level semantics and boundary-sensitive spatial structure need to be preserved during transfer. Representative examples include Channel-wise Distillation [26], which aligns category-sensitive channel responses between teacher and student, and Boundary Privileged Knowledge Distillation [27], which further separates body and edge supervision for more effective dense prediction.

In medical image segmentation, distillation has been widely adopted to improve efficiency while preserving strong segmentation performance. MSKD [28] transfers both semantic and structural information to enhance dense prediction quality. ACF-BCKD [29] further introduces attention-corrected feature transfer together with boundary-constrained supervision to better preserve fine spatial structure.

Recent studies have also investigated lightweight diffusion-based architectures for medical image segmentation. MedCondDiff [30] develops a semantically guided diffusion framework with reduced computational complexity, while MedSegDiffNCA [31] replaces conventional diffusion backbone components with neural cellular automata to construct a more efficient segmentation model. However, these methods primarily improve efficiency through architectural redesign rather than transferring knowledge from a pretrained diffusion teacher to a lightweight student. To the best of our knowledge, EFKD [16] is the first explicitly reported teacher-student distillation framework specifically developed for diffusion-based medical image segmentation. EFKD [16] transfers boundary-oriented knowledge from a large diffusion segmentor to a compact student model, enabling model compression while maintaining sensitivity to lesion contours. Nevertheless, its knowledge transfer is conducted predominantly in the spatial domain and does not explicitly account for the timestep-

dependent evolution of frequency information throughout the denoising process.

C. Frequency Distillation

Frequency-domain knowledge distillation extends conventional spatial supervision by transferring knowledge in the spectral domain. Wavelet Knowledge Distillation [32] uses wavelet decomposition to guide efficient image-to-image translation, while Multi-Scale Frequency-Driven Knowledge Distillation [33] performs spectral distillation across multiple scales. Frequency Attention for Knowledge Distillation [34] introduces frequency-domain attention to align student features with the teacher. Together, these studies show that frequency-aware supervision can preserve both global structure and fine detail more effectively than purely spatial matching.

However, most existing frequency distillation methods are developed for conventional recognition, translation, or dense prediction tasks, rather than diffusion-based generation. They typically treat supervision as static and do not consider how the reliability of spectral information changes during denoising. A recent Fourier-space analysis of DDPMs [15] shows that low-frequency content is recovered earlier, whereas high-frequency details are refined later in the reverse process. This spectral progression provides the main motivation for our work. Based on this view, Temporal Frequency Knowledge Distillation adapts spectral knowledge transfer to the denoising stage so that the student receives more suitable supervision throughout diffusion-based lesion segmentation.

III. PROPOSED METHOD

A. Teacher-Student Architecture

Our framework adopts a teacher-student architecture, as illustrated in Fig. 1. Specifically, we employ the original DermoSegDiff model [8] as the teacher, while LiteDermoSegDiff [16] serves as the student. In this way, the student preserves the same overall diffusion-based segmentation pipeline and encoder-decoder formulation, while using a more compact denoising backbone with reduced capacity.

During training, the teacher is kept frozen and provides supervision to the student. Both networks are trained using the same inputs at each corresponding timestep to ensure consistent guidance throughout the diffusion process. Since the student remains structurally aligned with the teacher, intermediate representations can be naturally distilled across corresponding stages. On top of the standard diffusion objective, we further introduce a timestep-conditioned frequency-domain distillation loss to transfer spectral knowledge from teacher to student in a manner consistent with the denoising process.

B. Feature Alignment

LiteDermoSegDiff [16] is obtained by simplifying the DermoSegDiff teacher [8] while preserving representative stages from the original diffusion-based segmentation pipeline. As a result, the teacher and student still produce feature maps at several compatible resolutions during denoising. We therefore

align their intermediate representations in a direct and stage-consistent manner, rather than introducing additional projection layers or learnable adapters.

Specifically, feature pairs are selected only when the teacher and student representations share the same spatial resolution and correspond to the same denoising timestep. This ensures that the spectral discrepancy is computed between semantically and structurally comparable features. Such a simple alignment strategy keeps the distillation process lightweight, avoids extra computational overhead, and allows the student to receive guidance from the retained structural stages of the teacher throughout the diffusion trajectory.

C. Temporal Frequency Knowledge Distillation

Let f_T^l and f_S^l denote an aligned teacher-student feature pair at level l , where only feature maps with matching spatial resolutions are used for distillation. Their spectral representation features F are obtained via the two-dimensional Fourier transform:

$$F_T^l = \mathcal{F}(f_T^l), \quad F_S^l = \mathcal{F}(f_S^l), \quad (1)$$

where $\mathcal{F}(\cdot)$ denotes the orthonormal FFT.

For each aligned feature level, we define a radial frequency map, denoted by r , over the FFT coordinate grid:

$$r^l(i, j) = \sqrt{f_y(i)^2 + f_x(j)^2}, \quad \tilde{r}^l(i, j) = \frac{r^l(i, j)}{\max_{i, j} r^l(i, j)}, \quad (2)$$

where i and j index the vertical and horizontal frequency bins, and (f_y, f_x) are the corresponding frequency coordinates along the two axes. The normalized radius $\tilde{r}^l(i, j) \in [0, 1]$ serves as a continuous indicator of spatial variation scale within the feature representation: values near zero correspond to more slowly varying coarse-scale feature patterns, whereas larger values correspond to more rapidly varying fine-scale patterns.

Following DDPM [6], $\beta_t \in (0, 1)$ denotes the variance schedule that controls the noise injected at timestep t , while $\alpha_t = 1 - \beta_t$ represents the corresponding signal retention factor. The cumulative retained signal is defined as $\bar{\alpha}_t = \prod_{k=1}^t \alpha_k$. As t increases, more noise is accumulated and $\bar{\alpha}_t$ decreases monotonically. Therefore, $\bar{\alpha}_t$ naturally indicates the denoising state, with larger values corresponding to cleaner states and smaller values to noisier ones.

Based on this observation, we define a spectral weight map as follows:

$$w(\tilde{r}, t) = 1 + \eta_h \bar{\alpha}_t^\gamma \tilde{r} + \eta_l (1 - \bar{\alpha}_t^\gamma)(1 - \tilde{r}), \quad (3)$$

where η_h controls the emphasis on higher-frequency components, η_l controls the low-frequency structural bias, and γ determines the timestep sensitivity of the high-frequency term. In this formulation, $\bar{\alpha}_t^\gamma$ acts as a high-frequency temporal gate, while $(1 - \bar{\alpha}_t^\gamma)$ modulates the low-frequency term. When the denoising state is relatively clean, more high-frequency guidance can be transferred; conversely, at noisier stages, the low-frequency structural guidance is strengthened to provide more stable supervision.

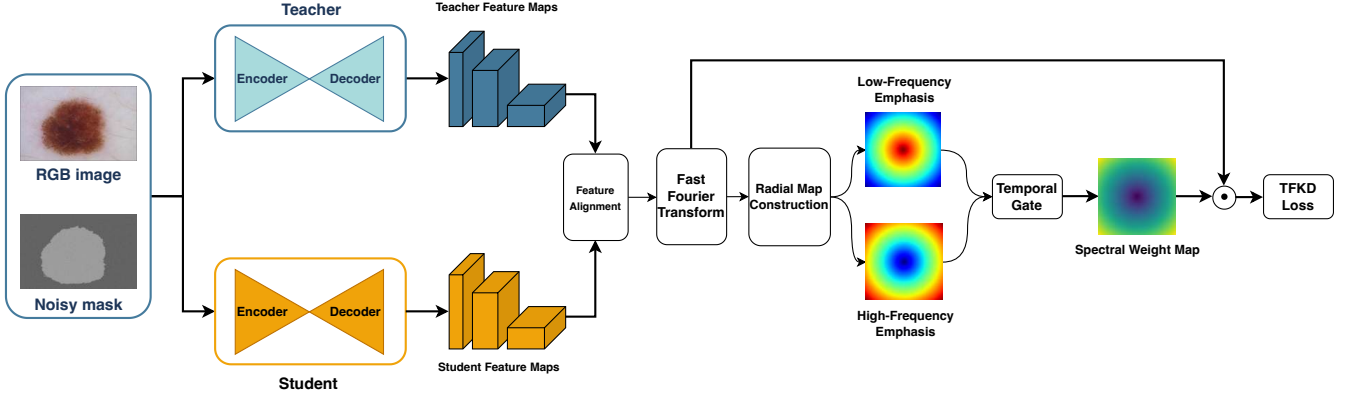


Fig. 1. Overview of the proposed **Temporal Frequency Knowledge Distillation (TFKD)** framework. The teacher and student are diffusion-based segmentation networks predicting aligned feature representations from the same RGB image, noisy mask, and timestep. These features are transformed into the frequency domain to compute the spectral difference between teacher and student. A timestep-conditioned spectral weight map is constructed by creating a 2D frequency grid, converting it into a radial map, decomposing low- and high-frequency components, and modulating them with a temporal gate. The weighted spectral discrepancy is then used to define the temporal spectral distillation loss for supervising the student.

TABLE I
QUANTITATIVE COMPARISON ON ISIC2018 [17].

Methods	Params↓	GFLOPs↓	FPS↑	DSC↑	SE↑	SP↑	ACC↑
U-Net [2]	31.0M	27.37	208.30	0.8729	0.8739	0.9515	0.9303
Swin-UNet [5]	41.3M	35.80	40.23	0.8532	0.8148	0.9641	0.9232
DeepLabv3+ [3]	41.1M	22.88	125.30	0.8820	0.8560	0.9770	0.9510
MissFormer [35]	42.4M	20.33	54.24	0.8697	0.8256	0.9725	0.9320
DermoSegDiff [8]	81.6M	1990	14.20	0.8926	0.8812	0.9647	0.9419
LiteDermoSegDiff [16]	9.3M	1865	36.43	0.8616	0.7904	0.9833	0.9304
KD [11]	9.3M	1865	36.43	0.8733	0.8465	0.9653	0.9327
EFKD [16]	9.3M	1865	36.43	0.8826	0.8715	0.9616	0.9365
TFKD (Ours)	9.3M	1865	36.43	0.8986	0.8855	0.9774	0.9582

Best and second-best metric values are highlighted in **blue** and **red**, respectively.

Using this weighting scheme, the proposed Temporal Frequency Knowledge Distillation loss is defined as:

$$\mathcal{L}_{\text{TFKD}} = \frac{1}{L} \sum_{l=1}^L \text{Mean} \left(w^l(t) \odot |F_S^l - F_T^l|^2 \right), \quad (4)$$

where $w^l(t)$ denotes the spectral weight map at level l , $\odot$ is element-wise multiplication, and L is the number of aligned feature levels. Here, F_S^l and F_T^l are the complex Fourier coefficients of the student and teacher features, respectively, while $\text{Mean}(\cdot)$ averages the weighted spectral discrepancy over all coefficients. By operating in the complex Fourier domain, the objective accounts for differences in both magnitude and phase. The final distillation term is obtained by averaging across all aligned levels.

Training procedure. The student network is optimized jointly using diffusion reconstruction and frequency-domain distillation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{TFKD}} \mathcal{L}_{\text{TFKD}}, \quad (5)$$

where λ_{TFKD} balances the distillation regularization against the original denoising objective.

IV. EXPERIMENTAL RESULTS

A. Implementation Details

Our method is implemented in PyTorch (v1.14.0) and trained on a single NVIDIA RTX 3090Ti GPU. We adopt a diffusion process with $t = 250$ steps and linearly increasing variances from $\beta_{\text{start}} = 0.0004$ to $\beta_{\text{end}} = 0.08$. Training runs for 2,000 epochs with batch size 32 using Adam optimizer at an initial learning rate of 1×10^{-4} , reduced by a factor of 0.5 when there is no improvement in the last 10 epochs. The TFKD loss weights are empirically set as $\lambda_{\text{TFKD}} = 1$, $\eta_h = 2$ and $\eta_l = 1$, with a timestep sensitivity parameter $\gamma = 0.8$. We augment data via Albumentations, including flipping, coarse dropout, and additional intensity/geometric perturbations. At inference, nine segmentation masks are sampled per test image, averaged, and thresholded to produce the final prediction.

TABLE II
QUANTITATIVE COMPARISON ON HAM10000 [18].

Methods	Params↓	GFLOPs↓	FPS↑	DSC↑	SE↑	SP↑	ACC↑
U-Net [2]	31.0M	27.37	208.30	0.9317	0.9480	0.9447	0.9480
Swin-UNet [5]	41.3M	35.80	40.23	0.9283	0.9336	0.9723	0.9656
DeepLabv3+ [3]	41.1M	22.88	125.30	0.9268	0.9277	0.9768	0.9651
MissFormer [35]	42.4M	20.33	54.24	0.9254	0.9271	0.9760	0.9644
DermoSegDiff [8]	40.3M	2556	17.12	0.9487	0.9351	0.9887	0.9759
LiteDermoSegDiff [16]	8.5M	2344	31.27	0.9402	0.9278	0.9795	0.9716
KD [11]	8.5M	2344	31.27	0.9446	0.9421	0.9802	0.9734
EFKD [16]	8.5M	2344	31.27	0.9453	0.9506	0.9810	0.9738
TFKD (Ours)	8.5M	2344	31.27	0.9522	0.9498	0.9863	0.9775

Best and second-best metric values are highlighted in blue and red, respectively.

B. Datasets

To evaluate the proposed method, we use two publicly available skin lesion segmentation datasets, namely ISIC2018 [17] and HAM10000 [18], together with the salient object detection dataset DUTS [19]. **ISIC2018** [17] contains 2,594 dermoscopic images with pixel-level annotations of skin lesions, split into 1,815/259/520 for train/val/test following the current repository setup. **HAM10000** [18] contains 10,015 dermoscopic images and their corresponding segmentation masks, split into 7,200/1,800/1,015 for train/val/test. **DUTS** [19] is a large-scale salient object detection benchmark containing natural images with pixel-level saliency annotations. In our experiments, it is used to evaluate the generalization ability of the proposed framework beyond dermoscopic lesion segmentation. Each image in all datasets is resized to 128×128 pixels.

C. Evaluation Metrics

To comprehensively evaluate segmentation performance, we report Dice Score (DSC), Sensitivity (SE), Specificity (SP), Accuracy (ACC), and mean Intersection over Union (mIoU). These metrics measure complementary aspects of segmentation quality. Let TP, TN, FP, and FN denote the numbers of true-positive, true-negative, false-positive, and false-negative pixels, respectively.

Dice Score (DSC) measures the overlap between the predicted mask and the ground-truth mask. It is the primary metric for lesion segmentation because it directly reflects region-level agreement, especially under foreground-background imbalance:

$$\text{DSC} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}. \quad (6)$$

Sensitivity (SE), also known as recall or true positive rate, measures the proportion of lesion pixels that are correctly identified. A higher SE indicates that fewer lesion regions are missed:

$$\text{SE} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (7)$$

Specificity (SP), also known as true negative rate, measures the proportion of background pixels that are correctly classified. It reflects the ability of a model to avoid false lesion predictions in healthy/background regions:

$$\text{SP} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \quad (8)$$

Accuracy (ACC) measures the overall proportion of correctly classified pixels, including both foreground and background classes:

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (9)$$

Mean Intersection over Union (mIoU) averages the intersection-over-union score across classes. It is used to assess general segmentation quality, particularly for the DUTS generalization experiment:

$$\text{mIoU} = \frac{1}{C} \sum_{c=1}^C \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad (10)$$

where C denotes the number of classes, and TP_c , FP_c , and FN_c are computed for class c .

D. Results and Discussion

1) *Quantitative Results on ISIC2018 [17]*: Table I summarizes the ISIC2018 results [17]. TFKD delivers the best scores in 3 major accuracy metrics, reaching 0.8986 DSC, 0.8855 SE, and 0.9582 ACC, while also obtaining the second-best SP of 0.9774. Compared with LiteDermoSegDiff [16], the gains are substantial: DSC increases from 0.8616 to 0.8986, SE from 0.7904 to 0.8855, and ACC from 0.9304 to 0.9582. This improvement shows that the proposed temporal-frequency supervision helps the compact student recover both lesion regions and boundary-sensitive pixels more effectively. Compared with EFKD [16], TFKD further improves DSC and ACC, indicating that timestep-adaptive spectral supervision provides more effective guidance than static spatial boundary-oriented distillation. More importantly, TFKD surpasses the DermoSegDiff teacher [8] on all 4 segmentation metrics, demonstrating that the distilled student can achieve stronger lesion delineation despite using a much smaller architecture.

TABLE III
QUANTITATIVE COMPARISON ON DUTS [19].

Methods	mIoU $\uparrow$	ACC $\uparrow$
DermoSegDiff (Teacher) [8]	0.6436	0.9327
LiteDermoSegDiff [16]	0.6385	0.9301
KD [11]	0.6238	0.9303
EFKD [16]	0.6400	0.9303
TFKD (Ours)	0.6417	0.9339

Best and second-best metric values are highlighted in **blue** and **red**, respectively.

2) *Quantitative Results on HAM10000 [18]*: Table II presents the results on HAM10000 [18]. On this larger dataset, TFKD again provides the strongest overall performance, achieving the best DSC of 0.9522 and the best ACC of 0.9775. It also reaches the second-best SE and SP, with only marginal gaps to the top values. Relative to LiteDermoSegDiff [16], TFKD improves all segmentation metrics, confirming that the proposed distillation objective is not overfitted to ISIC2018 [17] and remains effective under a larger and more diverse dermoscopic distribution. Compared with EFKD [16], TFKD improves DSC, SP, and ACC while preserving almost the same sensitivity. In addition, TFKD exceeds the DermoSegDiff teacher [8] in DSC, SE, and ACC. These consistent gains suggest that temporal-frequency distillation transfers useful structural and boundary knowledge more effectively than conventional output-level or static frequency-based distillation.

3) *Quantitative Results on DUTS [19]*: Table III evaluates cross-domain generalization on DUTS [19]. Without changing the configuration used for ISIC2018 [17], TFKD obtains 0.6417 mIoU and 0.9339 ACC. It outperforms LiteDermoSegDiff [16], KD [11], and EFKD [16] on both metrics, showing that the learned temporal-frequency knowledge is not limited to dermoscopic images. Although the teacher achieves a slightly higher mIoU, TFKD obtains the best ACC among all methods and remains very close to the teacher in region overlap. These results show that the proposed distillation strategy remains effective when applied to a non-dermoscopic segmentation dataset.

Efficiency Trade-off Analysis. The efficiency-related metrics further highlight the practical value of TFKD. In terms of parameters, TFKD keeps the same compact student size as LiteDermoSegDiff [16], using only 9.3M parameters on ISIC2018 [17] and 8.5M parameters on HAM10000 [18], which is much smaller than the corresponding DermoSegDiff teacher [8]. In terms of FPS, TFKD also preserves the student inference speed and is clearly faster than the teacher. The only less favorable metric is GFLOPs: on ISIC2018 [17], TFKD still requires 1865 GFLOPs, and on HAM10000 [18] it requires 2344 GFLOPs. This is because diffusion-based segmentation performs iterative denoising rather than direct single-step prediction [6], [10]. Therefore, the reported GFLOPs primarily arise from the repeated denoising op-

TABLE IV
ABLATION STUDY OF OUR TFKD COMPONENTS ON LITEDERMOSEGDIFF BASELINE ON ISIC2018 [17].

Model \ Components	Base	HF	LF	TG	DSC $\uparrow$
	✓				0.8733
LiteDermoSegDiff	✓	✓			0.8846
	✓	✓	✓		0.8901
	✓	✓	✓	✓	0.8986

Base: Base term, HF: High-frequency
LF: Low-frequency, TG: Timestep gating

erations inherent to diffusion-based inference. Nevertheless, TFKD preserves the compact parameter count and inference speed of the student model while remaining faster than the diffusion teacher.

Qualitative Results. Qualitative comparisons in Fig. 2 visually validate the effectiveness of our proposed distillation strategy. As observed, the plain LiteDermoSegDiff [16] student and baseline distillation methods (KD [11] and EFKD [16]) often struggle with ambiguous regions, resulting in predicted blue contours that noticeably deviate from the green ground-truth. In contrast, the student model trained with TFKD produces much more coherent lesion boundaries that align better with the actual contours. While there are still some visible differences compared to the large DermoSegDiff [8] teacher, our TFKD-distilled model recovers a substantial portion of the teacher’s boundary awareness. Despite its compact capacity, the proposed method effectively bridges the visual performance gap, yielding reliable boundary representations that are highly valuable for clinical diagnosis.

V. ABLATION STUDY

Ablation on TFKD components. To isolate the contribution of each component in the updated TFKD design, we conduct a progressive ablation study on ISIC2018 [17]. The base variant performs frequency-domain feature alignment without explicit spectral preference, meaning that all Fourier coefficients are treated uniformly. We then incrementally introduce high-frequency (HF), low-frequency (LF), and timestep gating (TG). These components correspond to the main design choices of TFKD: emphasizing boundary-sensitive details, preserving coarse lesion structure, and adapting the spectral weights according to the diffusion denoising state.

Table IV shows a monotonic improvement as the proposed components are added. Starting from the base variant, the DSC is 0.8733. Adding HF increases the DSC to 0.8846, indicating that high-frequency supervision helps transfer fine boundary information from the teacher to the compact student. Introducing LF further improves the DSC to 0.8901, suggesting that low-frequency emphasis provides complementary global lesion guidance. Finally, adding TG achieves the best DSC of 0.8986. This last gain shows that timestep-aware modulation is useful because the importance of low- and high-frequency information changes across the diffusion trajectory.

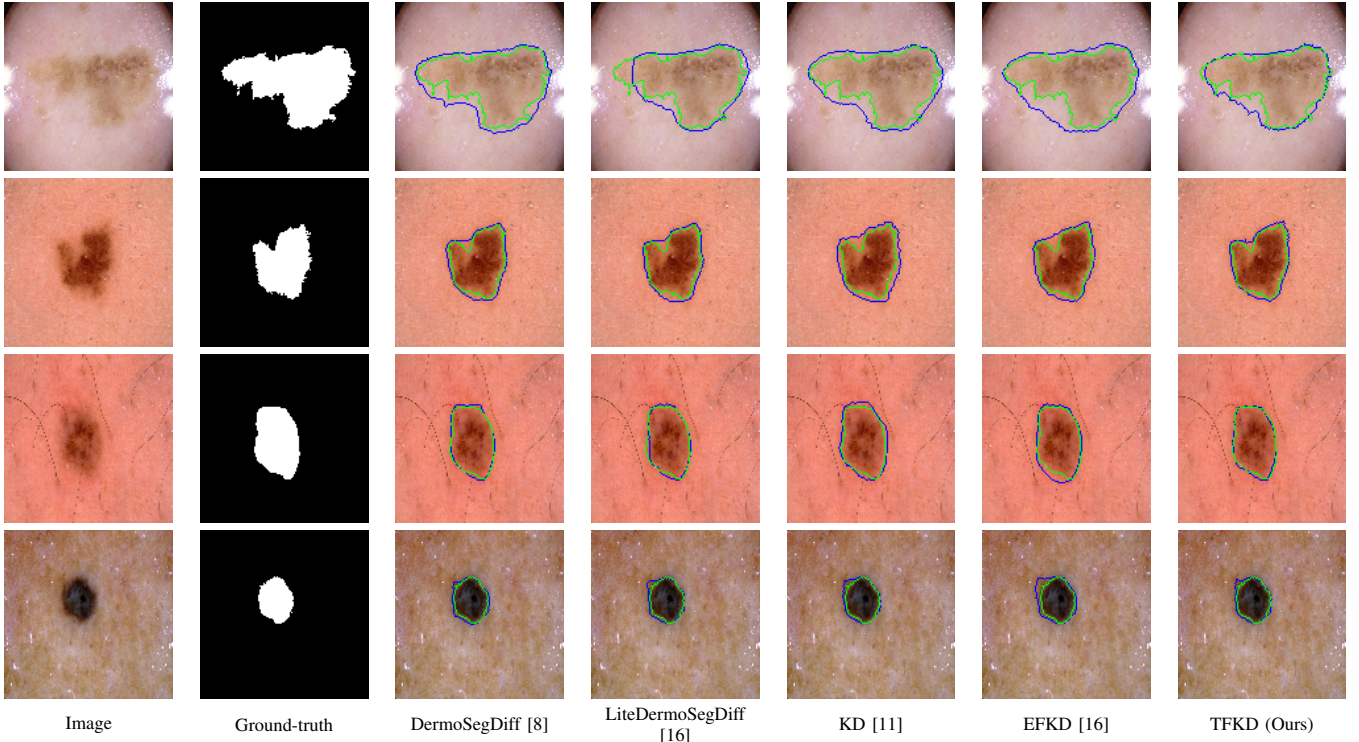


Fig. 2. Qualitative comparison of segmentation results on the ISIC2018 [17]. Ground-truth boundaries are shown in green; predicted boundaries are in blue.

Overall, the proposed TFKD formulation improves the DSC by 0.0253 over the base variant. This steady gain supports our central claim that effective distillation for diffusion-based segmentation should not rely on static spectral matching alone. Instead, combining high-frequency detail transfer, low-frequency structural stabilization, and timestep-aware frequency control provides a more suitable supervision signal for the student throughout the denoising process.

Hyperparameter sensitivity. TFKD balances spectral detail and structural guidance through η_h and η_l , while γ adjusts this balance along the diffusion trajectory. Increasing η_h strengthens the transfer of fine boundary information, whereas increasing η_l places greater emphasis on coarse lesion structure. Neither contribution should dominate, since accurate segmentation requires both well-defined boundaries and consistent lesion regions. Timestep gating γ determines how quickly high-frequency emphasis decreases as the noise level rises. The illustrative curves in Fig. 3 peak at $\eta_h = 2$, $\eta_l = 1$, and $\gamma = 0.8$ in their respective sweeps, with performance gradually declining beyond these values. This pattern suggests that balanced spectral supervision and a gradual timestep-dependent transition are preferable to aggressive weighting of either frequency range.

Limitations. Despite its effectiveness, the proposed TFKD framework still has several limitations. First, the spectral weighting mechanism relies on fixed hyperparameters (η_h , η_l , and γ), which may not be optimal for all lesion cases or datasets with different boundary complexity and frequency distributions. Second, the ablation focuses on DSC on ISIC2018

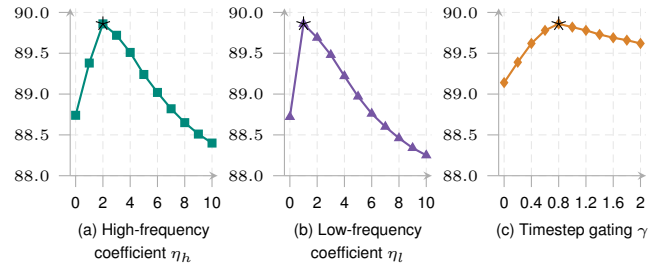


Fig. 3. Impact of the (a) high-frequency coefficient η_h , (b) low-frequency coefficient η_l , and (c) timestep gating γ on ISIC2018 [17].

[17]; broader validation with more datasets and additional metrics would provide a more complete understanding of each component. Third, as in other knowledge distillation methods, the student remains partially constrained by the teacher. If the teacher produces inaccurate high-frequency details or unreliable boundaries in ambiguous samples, these errors may also be transferred to the student. Future work will therefore explore adaptive spectral weighting and more robust teacher-student supervision to further improve generalization.

VI. CONCLUSION

In this paper, we propose Temporal Frequency Knowledge Distillation (TFKD), a timestep-aware frequency-domain distillation strategy for improving lightweight diffusion-based skin lesion segmentation. The proposed method exploits the spectral progression of diffusion denoising to adapt knowledge transfer across different timesteps, enabling the student

model to better learn both coarse lesion structure and fine boundary details. Built upon this idea, we improve LiteDermoSegDiff [16] and achieve consistent performance gains on the ISIC2018 [17] and HAM10000 [18] benchmarks. Furthermore, we demonstrate the effectiveness of the proposed distillation strategy in a general setting on DUTS [19]. These results show that TFKD is a simple yet effective way to enhance compact diffusion-based segmentation models without compromising inference efficiency.

Moving forward, we plan to extend the proposed framework to broader dense prediction tasks and investigate stronger general adaptation settings. Most notably, we aim to adapt the temporal-frequency distillation mechanism for 3D medical image segmentation (e.g., MRI and CT volumetric data), where balancing global anatomical structures and fine-grained tissue boundaries across complex spatial dimensions remains a critical and computationally demanding challenge.

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